TV-SIDARHE Model

A Compartmental Model of Infectious Disease Transmission

Language: Python
Application Domain: Mathematical Epidemiology
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1. Introduction

This document describes the TV-SIDARHE model (Time-Varying SIDARHE), a novel epidemic model proposed as a structured refinement of the eight-compartment SIDARTHE framework originally developed by Giordano et al. (2020). The TV-SIDARHE model introduces two substantive mathematical innovations designed to improve the realism of epidemic simulation over longer time horizons.

First, the model removes the Threatened (T) compartment, yielding a streamlined seven-compartment architecture (S, I, D, A, R, H, E). This reduction simplifies the parameter space while retaining the core detection asymmetry and severity stratification of the original model.

Second, the TV-SIDARHE model introduces two time-dependent functional forms absent from classical compartmental models: a logarithmically growing detection rate governed by epidemic detection capacity (parameter a), and an exponentially decaying transmission rate governed by government intervention intensity (parameter b). These mechanisms produce a dynamically adaptive model in which transmission and detection evolve endogenously with the time index, without requiring external forcing terms.

2. Compartmental Structure

2.1 The Seven Compartments

The TV-SIDARHE model partitions the normalised population N = 1 into seven states, six of which are dynamic and one of which (E) is a strictly absorbing mortality sink. The conservation identity holds at all time steps:

$$S(t) + I(t) + D(t) + A(t) + R(t) + H(t) + E(t) = N, \ \forall t$$

Symbol	Full Name	Epidemiological Role	Maps from SIDARTHE
S	Susceptible	Uninfected individuals; depleted by time-decaying force of infection from I and A	S (unchanged)
I	Infected (undetected)	Infected but undiagnosed; principal driver of transmission; subject to logarithmically growing diagnosis rate	I (unchanged)
D	Diagnosed	Confirmed positive cases; reduced transmission; may progress to Recognised	D (unchanged)
A	Ailing (undetected)	Symptomatic and undiagnosed; contributes to transmission; subject to logarithmically growing recognition rate	A (unchanged)
R	Recognised	Formally managed symptomatic cases; the sole source of disease mortality in this model	R (merged with T)
H	Healed	Cumulative recoveries from all infected compartments; absorbing healthy state	H (unchanged)
E	Extinct	Cumulative disease mortality; absorbing death state; fed exclusively from R	E (re-routed from T)

2.2 Structural Comparison with Original SIDARTHE

Feature	SIDARTHE (Giordano et al.)	TV-SIDARHE (This Model)
Compartments	8: S, I, D, A, R, T, H, E	7: S, I, D, A, R, H, E
ICU compartment (T)	Present; dedicated severe-case pathway	Removed; flows absorbed into R
Mortality source	T compartment only (τ)	R compartment only (μ)
Transmission rates	Constant β1, β2, β3, β4	Time-decaying: $b^{-I}\cdot\alpha$ and $b^{-I}\cdot\beta$; b encodes government intervention intensity
Detection rates	Constant α1 (I→D), α2 (A→R)	Logarithmically growing: $(\log_a(t+\delta)+1)\cdot\varepsilon$ and $(\log_a(t+\delta)+1)\cdot\theta$; a encodes epidemic detection capacity
Free parameters	25 (incl. T vector)	22 (incl. T vector)
Transmission compartments	I, D, A, R (four sources)	I, A (two sources)

3. Mathematical Formulation

3.1 Time-Dependent Functional Forms

Two time-dependent modifiers are central to the TV-SIDARHE framework. Both are functions of the discrete time index i. Parameter b encodes government intervention intensity and governs transmission decay; parameter a encodes epidemic detection capacity and governs detection growth.

3.1.1 Government Intervention: Exponential Transmission Decay

Government intervention intensity is represented by parameter b > 1. The transmission force from both I and A is multiplied by the factor b^{-I}, producing a monotonically decreasing effective transmission rate. Larger values of b correspond to stronger or more rapidly effective interventions; values of b close to 1 represent weak or absent intervention.

$$\Lambda(t) = b^{-I} \cdot [\alpha \cdot I(t) + \beta \cdot A(t)]$$

In the reference run, b = 1.05. At t = 150, the decay factor is $1.05^{-150} \approx 0.0007$, meaning effective transmission is reduced to less than 0.1% of its baseline value at time step 150.

3.1.2 Epidemic Detection Capacity: Logarithmic Detection Growth

Epidemic detection capacity is represented by parameter a > 1. Detection and recognition rates grow over time according to the factor $(\log_a(i + \delta) + 1)$, where $\delta = 10^{-10}$ prevents a domain error at i = 0. Larger values of a correspond to faster-expanding detection capacity; the logarithmic form produces sub-linear growth, so the modifier increases quickly at small time indices and more slowly at larger ones.

$$\Phi(t) = \log_a(t + \delta) + 1$$

The constant +1 ensures the modifier equals 1 at t = 0 (since $\log_a(\delta) \approx 0$ for small δ), preserving the baseline detection parameter values at the start of the simulation.

In the reference run, a = 2. At t = 100, the detection multiplier is $\log_2(100 + \delta) + 1 \approx 7.64$, meaning diagnosis and recognition rates are approximately 7.6× their baseline values at time step 100.

3.2 Complete Difference Equations

Applying both time-dependent modifiers, the full forward Euler system governing the TV-SIDARHE model is:

```
# Susceptible: depleted by time-decaying force of infection (I and A only)
S[t+1] = S[t] - b⁻ᴵ·α·I[t] - b⁻ᴵ·β·A[t]

# Infected (undetected): gain from transmission; lose to diagnosis, ailing, natural
recovery
```

```
I[t+1] = I[t] + b⁻ᴵ·α·I[t] + b⁻ᴵ·β·A[t] - [Φ(t)·ε + ζ + ρ]·I[t]

# Diagnosed: gain from I via growing diagnosis rate; lose to recognition and recovery
D[t+1] = D[t] + Φ(t)·ε·I[t] - (ζ + λ)·D[t]

# Ailing (undetected): gain from symptom onset in I; lose to growing recognition and
recovery
A[t+1] = A[t] + ζ·I[t] - [Φ(t)·θ + σ]·A[t]

# Recognised: gain from D (ζ) and A via growing recognition; lose to mortality and
recovery
R[t+1] = R[t] + ζ·D[t] + Φ(t)·θ·A[t] - (κ + μ)·R[t]

# Healed: cumulative recoveries from I, D, A, R
H[t+1] = H[t] + ρ·I[t] + λ·D[t] + σ·A[t] + κ·R[t]

# Extinct: cumulative mortality from R only
E[t+1] = E[t] + μ·R[t]

# Where: Φ(t) = log_a(t + 10⁻¹⁰) + 1  (detection capacity growth; base a)
#         b⁻ᴵ                              (government intervention decay; base b)
```

3.3 Non-Negativity Guard

As in the original SIDARTHE implementation, the TV-SIDARHE model includes a non-negativity guard on the susceptible compartment. At each time step, the model checks whether the time-decaying force of infection would exceed the remaining susceptible pool:

$$\text{if } S[t] < b^{-I} \cdot \alpha \cdot I[t] + b^{-I} \cdot \beta \cdot A[t]:$$

If this condition is met, `S[t+1]` is clamped to zero and the new infection term in the I equation uses the residual value `S[i]` (the remaining susceptibles before exhaustion). All other compartment equations proceed identically in both branches. Because the transmission force b^{-I} decays with time as government intervention accumulates, this condition is most likely to trigger in the early simulation phase when b^{-I} is close to 1.

4. Model Parameters

4.1 Transmission Parameters

Symbol	Code	Role	Interpretation	Ref. Value
α	α	I → force of infection	Baseline transmission rate from I; effective rate at step i is $b^{-I}\cdot\alpha$	0.8
β	β	A → force of infection	Baseline transmission rate from A; effective rate at step i is $b^{-I}\cdot\beta$	0.8

Symbol	Code	Role	Interpretation	Ref. Value
b	`b`	Government intervention intensity	Base of the exponential decay b^{-I} applied to transmission; b > 1 ensures monotone decay; larger b represents stronger intervention	1.05
a	`a`	Epidemic detection capacity	Base of the logarithmic growth factor Φ(t); controls the rate of detection expansion; larger a represents faster-growing detection capacity	2

4.2 Disease Progression Parameters

Symbol	Code	Transition	Interpretation	Ref. Value	Mean Duration (time steps)
ε	`ε`	I → D	Baseline diagnosis rate; effective rate at step t is Φ(t)·ε	0.016	1/0.016 = 62.5 (baseline)
ζ	`ζ`	I → A and D → R	Rate of symptom onset (I→A) and progression from diagnosed to recognised (D→R)	0.07	1/0.07 ≈ 14.29
θ	`θ`	A → R	Baseline recognition rate; effective rate at step t is Φ(t)·θ	0.0156	1/0.0156 ≈ 64.10 (baseline)

4.3 Recovery and Mortality Parameters

Symbol	Code	Source	Interpretation	Ref. Value
ρ	`ρ`	I	Natural recovery rate from undetected infection (mild course)	0.55
λ	`λ`	D	Recovery rate of diagnosed individuals under monitoring	0.023
σ	`σ`	A	Recovery rate of ailing (symptomatic, undetected) patients	0.0168
κ	`κ`	R	Recovery rate of formally recognised patients	0.14
μ	`μ`	R	Mortality rate; sole death source in TV-SIDARHE	0.1238

4.4 Complete Reference Instantiation

```
s = adjust_SIDARTHE(
    T,                          # Time vector: [0, 1, ..., 149]
```

```
    N   = 1,                    # Normalised population
    I   = 0.001,                # Initial infected (undetected): 0.1%
    D   = 0,                    # Initial diagnosed: 0
    A   = 0.001,                # Initial ailing (undetected): 0.1%
    R   = 0,                    # Initial recognised: 0
    H   = 0,                    # Initial healed: 0
    E   = 0,                    # Initial extinct: 0
    α   = 0.8,                   # Baseline transmission rate (I)
    β   = 0.8,                   # Baseline transmission rate (A)
    ε   = 0.016,                # Baseline diagnosis rate (I→D)
    ζ   = 0.07,                 # Symptom onset / D→R progression rate
    θ   = 1.56e-02,             # Baseline recognition rate (A→R)
    ρ   = 0.55,                 # Recovery rate (I)
    σ   = 1.68e-02,             # Recovery rate (A)
    κ   = 0.023,                # Recovery rate (R)
    λ   = 0.14,                 # Recovery rate (D)
    μ   = 0.1238,               # Mortality rate (R→E)
    a   = 2,                    # Epidemic detection capacity (logarithm base)
    b   = 1.05                  # Government intervention intensity (decay base)
)
```

5. Class Architecture

5.1 Class Overview

The model is implemented as a Python class named `adjust_SIDARTHE`. In the model family this class is referred to as TV-SIDARHE. The class follows the same three-method interface established in the companion SEIR and SIDARTHE implementations.

```
class adjust_SIDARTHE:   # TV-SIDARHE
    def __init__(self, T, N, I, D, A, R, H, E, α, β, ε, ζ, θ, ρ, σ, κ, λ, μ, a, b)
    def calc(self)    # Time-dependent numerical integration with non-negativity guard
    def plot(self)    # Seven-curve visualisation via Matplotlib
```

5.2 Constructor: __init__

The constructor accepts 22 arguments and stores them as instance attributes. Seven compartment histories are initialised as single-element lists. The initial susceptible fraction is computed as the complement of all other compartments, fully satisfying the conservation identity:

```
self.S = [N - I - D - A - R - H - E]
# With reference values: S₀ = 1 - 0.001 - 0 - 0.001 - 0 - 0 - 0 = 0.998
```

The initial susceptible fraction is computed as $S_0 = N - I_0 - D_0 - A_0 - R_0 - H_0 - E_0$, fully satisfying the conservation identity for arbitrary initial conditions. With the reference values, $S_0 = 1 - 0.001 - 0.001 = 0.998$.

5.3 Computational Method: calc()

5.3.1 Idempotency Guard

The method opens with the check `if len(self.T) == len(self.S)`, which returns immediately if the simulation has already been completed. This prevents list over-extension on repeated calls and makes the method safe to call multiple times.

5.3.2 Time-Index Computation

At each iteration, both time-dependent modifiers are recomputed using the current loop index i. The Python `math.log(i + 0.1**10, self.a)` call evaluates the logarithm of $i + 10^{-10}$ in base a. The offset `0.1**10 = 10⁻¹⁰` is computationally negligible (approximately 10^{-10}) but prevents a `ValueError` at i = 0 where log(0) would be undefined. The +1 addend ensures the modifier equals 1.0 at the limit $i \to 0$.

5.3.3 Branch Structure

The conditional non-negativity guard evaluates the time-decaying force of infection at each step and selects between two branches:

```
for i in range(0, len(self.T) - 1):
    Φ = math.log(i + 0.1**10, self.a) + 1      # Detection capacity growth factor
(base a)
    Λ = self.b**(-i) * (self.alpha*self.I[i] + self.beta*self.A[i])  # Force of
infection

    if self.S[i] < Λ:           # Guard: susceptible pool would be exhausted
        self.S.append(0)         # Clamp S to zero
        self.I.append(self.I[i] + self.S[i] - (Φ*self.epsilon + self.zeta +
self.rho)*self.I[i])
    else:                        # Normal update
        self.S.append(self.S[i] - Λ)
        self.I.append(self.I[i] + Λ - (Φ*self.epsilon + self.zeta +
self.rho)*self.I[i])
    # D, A, R, H, E equations are identical in both branches
```

5.4 Visualisation Method: plot()

The `plot()` method renders seven time-series curves. It follows the same lazy-evaluation pattern: if `len(self.S) == 1`, then `calc()` is invoked automatically before rendering. The Threatened (T) curve present in the original SIDARTHE plot is absent here, reflecting the removal of that compartment.

Colour	Compartment	Expected Trajectory
Blue	**Susceptible (S)**	Sigmoidal decrease from ~0.998; decay rate driven by b^{-I} (government intervention intensity), accelerating early then slowing as the intervention effect accumulates.
Red	**Infected (I)**	Unimodal peak representing the undetected infectious wave; peak height and timing sensitive to α and b.
Yellow	**Diagnosed (D)**	Lagged unimodal peak following I; grows as Φ(t)·ε rises but declines as I is depleted.

Colour	Compartment	Expected Trajectory
Cyan	**Ailing (A)**	Symptomatic undetected; peak between I and R; shape influenced by logarithmic recognition growth.
Magenta	**Recognised (R)**	Formally managed cases; key driver of both recoveries (κ) and mortality (μ); peaks later than A.
Black	**Healed (H)**	Monotonically increasing; dominant final state; accumulates recoveries from all four infected compartments.
Orange	**Extinct (E)**	Monotonically increasing from zero; magnitude determined by cumulative R and mortality rate μ.

6. Comparative Summary Across the Model Family

Feature	SEIR	SIDARTHE	TV-SIDARHE (This Model)
Compartments	4	8	7
Free parameters	10	25	22
Detection modelling	None	Fixed rates	Logarithmically growing rates
Transmission over time	Constant	Constant	Exponentially decaying
ICU dynamics	No	Yes (T compartment)	No (T removed)
Mortality pathway	None	T only	R only
Biological complexity	Low	High	Medium-High
Calibration difficulty	Low	High	Moderate
Primary novelty	—	Detection stratification	Government intervention (b) and detection capacity (a) as time-varying parameters

7. Summary

The TV-SIDARHE model (Time-Varying SIDARHE) represents an original contribution to the family of compartmental epidemic models. Government intervention intensity is encoded by parameter b, which drives exponential transmission decay b^{-I}; epidemic detection capacity is encoded by parameter a, which drives logarithmic detection growth ($\log_a(t + \delta) + 1$). Together, these two mechanisms allow transmission and detection to evolve endogenously with the time index without requiring additional external parameters.

Relative to the original SIDARTHE model, TV-SIDARHE achieves a more parsimonious structure by

removing the Threatened (T) compartment and routing the mortality pathway through the Recognised (R) state. This reduces the parameter burden while preserving the core detection-stratified architecture. The three-method class interface maintains consistency with the companion SEIR and SIDARTHE implementations.

References

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